

Open-system model for light-shift-driven multiparameter spin metrology

1 Setup

We consider a single spin J (one atom of a product ensemble of N_a atoms; for $J = 3$ this is the Cs $F = 3$ hyperfine manifold) evolving under a closed-system Hamiltonian that is the sum of a static field to be estimated, an rf control rotation with a piecewise-constant random phase, and a nonlinear term generated by the tensor light shift of an off-resonant, linearly polarized beam. With $\hbar = 1$ and time in units of $1/\Omega$,

$$H(t) = \underbrace{\mathbf{w} \cdot \mathbf{J}}_{\text{field to estimate}} + \underbrace{\Omega [\cos \phi(t) J_x + \sin \phi(t) J_y]}_{\text{rf control}} + \underbrace{\frac{\beta}{2J} J_z^2}_{\text{tensor light shift}}, \quad (1)$$

where $\mathbf{w} = (w_x, w_y, w_z)$ is the three-component parameter vector and $\phi(t)$ is constant on blocks of `random_phase_steps` time steps and drawn uniformly on $[0, 2\pi)$ for every realization. The code's parameter `beta` enters as $\beta/2J$, as written.

2 Light-induced decoherence

The same beam that produces the J_z^2 term scatters photons. In the far-detuned limit (detuning large compared with the excited-state hyperfine splitting Δ'_{HF} but small compared with the fine-structure splitting) the scattering amplitude acts on the electron spin alone. For linear polarization the no-spin-flip (Rayleigh) part is a scalar and does not decohere the ground manifold. The spin-flip (Raman) part, projected onto the F manifold, is proportional to $J_{\pm}$ by the Wigner–Eckart theorem. Neglecting Raman loss to the other hyperfine manifold and any decoherence from a separate readout beam, the single decoherence channel is therefore the pair of jump operators

$$L_+ = \sqrt{\gamma} J_+, \quad L_- = \sqrt{\gamma} J_-, \quad J_{\pm} = J_x \pm iJ_y. \quad (2)$$

Both the tensor light shift and the scattering rate scale with intensity over detuning squared, $\beta \propto I \Delta'_{\text{HF}}/\Delta^2$ and $\gamma_s \propto I \Gamma/\Delta^2$, so their ratio is a property of the atom and the transition, not of the beam. We parametrize this by a fixed figure of merit

$$\eta \equiv \frac{\beta}{\gamma}, \quad \gamma = \frac{\beta}{\eta}, \quad \eta \sim c \frac{\Delta'_{\text{HF}}}{\Gamma} = \mathcal{O}(10\text{--}100), \quad (3)$$

with c an order-one Clebsch–Gordan factor obtainable from the ground-state polarizability and optical-pumping formalism of Deutsch and Jessen (Opt. Commun. **283**, 681 (2010)). Consequently the nonlinearity cannot be increased without increasing the decoherence in proportion, and a sweep over β is simultaneously a sweep over γ .

3 Master equation (Schrödinger picture)

The Lindblad master equation for the single-atom density matrix is

$$\frac{d\rho}{dt} = -i[H(t), \rho] + \gamma \left(J_+ \rho J_- - \frac{1}{2} \{J_- J_+, \rho\} \right) + \gamma \left(J_- \rho J_+ - \frac{1}{2} \{J_+ J_-, \rho\} \right). \quad (4)$$

Writing $J_{\pm} = J_x \pm iJ_y$, the cross terms cancel and the dissipator collapses to

$$\mathcal{D}[\rho] = 2\gamma \sum_{k=x,y} \left(J_k \rho J_k - \frac{1}{2} \{J_k^2, \rho\} \right) = -\gamma \left([J_x, [J_x, \rho]] + [J_y, [J_y, \rho]] \right), \quad (5)$$

a double-commutator (dephasing-type) channel along x and y , each at rate 2γ . It commutes with rotations about z , conserves the trace, and is unital: the maximally mixed state $\mathbb{K}/(2J+1)$ is its fixed point. Here $J_- J_+ + J_+ J_- = 2(J_x^2 + J_y^2) = 2[J(J+1) - J_z^2]$.

4 Master equation (Heisenberg picture)

Expectation values can equally be computed by propagating the observable with the adjoint generator,

$$\langle O \rangle(t) = \text{Tr}[O e^{t\mathcal{L}}(\rho_0)] = \text{Tr}[e^{t\mathcal{L}^\dagger}(O) \rho_0], \quad (6)$$

with

$$\begin{aligned} \frac{dO}{dt} &= i[H(t), O] + \gamma \left(J_- O J_+ - \frac{1}{2} \{J_- J_+, O\} \right) + \gamma \left(J_+ O J_- - \frac{1}{2} \{J_+ J_-, O\} \right) \\ &= i[H(t), O] - \gamma \left([J_x, [J_x, O]] + [J_y, [J_y, O]] \right). \end{aligned} \quad (7)$$

The dissipator takes the same form in both pictures because the jump set $\{J_+, J_-\}$ is closed under Hermitian conjugation. This is the form integrated in the simulation: the observable $O = J_y$ (or J_z) is propagated and the signal is $\text{Tr}[O(t)\rho_0]$ with ρ_0 the initial spin coherent state.

Exact relaxation of the first moments

Because $\mathcal{D}^\dagger$ maps the vector operator $\mathbf{J}$ onto itself,

$$\mathcal{D}^\dagger[J_x] = -\gamma J_x, \quad \mathcal{D}^\dagger[J_y] = -\gamma J_y, \quad \mathcal{D}^\dagger[J_z] = -2\gamma J_z, \quad (8)$$

so for any J the Bloch vector obeys

$$\frac{d}{dt} \langle J_x \rangle = (\text{coherent}) - \gamma \langle J_x \rangle, \quad (9)$$

$$\frac{d}{dt} \langle J_y \rangle = (\text{coherent}) - \gamma \langle J_y \rangle, \quad (10)$$

$$\frac{d}{dt} \langle J_z \rangle = (\text{coherent}) - 2\gamma \langle J_z \rangle. \quad (11)$$

The coherent part is not closed at first order because the J_z^2 term couples $\langle \mathbf{J} \rangle$ to second moments; it is handled numerically below. The parameter γ is therefore the transverse relaxation rate, and 2γ the longitudinal one.

5 Measurement model

At each time step n the recorded datum is

$$M_n = \langle O \rangle_n + \xi_n, \quad \xi_n \sim \mathcal{N}(0, \sigma^2), \quad \sigma = \frac{1}{\sqrt{N_p} G N_a}, \quad (12)$$

i.e. the ensemble-averaged single-atom expectation value with Gaussian photon shot noise from a separate readout beam of N_p photons per step, Faraday gain G , and N_a atoms. Back-action of the readout beam is not included. The estimator is the least-squares fit of $\mathbf{w}$ to $\{M_n\}$ using the same open-system forward model, with γ treated as known (it is fixed by the chosen β), and its covariance is compared with the classical Cramér–Rao bound $\text{Cov}(\hat{\mathbf{w}}) \geq F^{-1}$,

$$F_{ab} = \frac{1}{\sigma^2} \sum_n \frac{\partial \langle O \rangle_n}{\partial w_a} \frac{\partial \langle O \rangle_n}{\partial w_b}. \quad (13)$$

6 Numerical integration

Time is discretized with step δt . With $U_w = e^{-i\delta t \mathbf{w} \cdot \mathbf{J}}$ and $U_c[n] = e^{-i\delta t H_c[n]}$ the control step for block n , define $U_n = U_c[n] U_w$. The Hamiltonian and dissipative parts are split to first order,

$$\text{Schrödinger:} \quad \rho_{n+1} = e^{\delta t \mathcal{D}} (U_n \rho_n U_n^\dagger), \quad (14)$$

$$\text{Heisenberg:} \quad O_{n+1} = U_n^\dagger e^{\delta t \mathcal{D}^\dagger} (O_n) U_n. \quad (15)$$

The map $A \equiv e^{\delta t \mathcal{D}^\dagger}$ is built once per realization as a $(2J+1)^2 \times (2J+1)^2$ matrix acting on the vectorized observable (the exact solution of the dissipative part over one step), so no approximation beyond the first-order splitting is made. That splitting is of the same order as the pre-existing split between U_w and U_c . Because A is linear and independent of $\mathbf{w}$, the Fréchet-derivative recursion for $\partial O_n / \partial w_a$, needed for the Fisher matrix, is unchanged apart from passing O_n and $\partial O_n / \partial w_a$ through A first. Setting $\gamma = 0$ (no `--eta`) recovers the closed-system code exactly.

Expected consequence. The signal derivatives $\partial \langle O \rangle_n / \partial w_a$ decay on the time scale $1/\gamma$, so the accumulated Fisher information saturates near $T \sim 1/\gamma$ instead of growing polynomially. At a fixed total time T the useful nonlinearity is bounded by $\beta \lesssim \eta/T$, and the question becomes whether the nonlinearity improves multiparameter estimation enough to pay for the decoherence it necessarily brings.

7 Assumptions and how to relax them

The model above rests on two simplifications, adopted deliberately for the first study. Both can be removed within the same numerical scheme.

A1: the readout beam does not decohere the atoms

Relaxed in Section 8 (script v6).

The measurement model treats the N_p readout photons per step as pure shot noise. A readout beam scatters photons too, through the same Raman channel (2), so it adds to γ a rate γ_r set by the readout intensity rather than by β . Because Faraday signal and scattering both scale as

$1/\Delta^2$, the ratio of measurement signal-to-noise to scattering is independent of detuning and fixed by the resonant optical depth $\text{OD} = N_a \sigma_0 / A$: up to order-one polarization and branching factors, the number of photons scattered per atom over the whole run is

$$P_{\text{sc}} \simeq \frac{\text{SNR}_{\text{tot}}^2}{N_a \text{OD}}, \quad \text{SNR}_{\text{tot}}^2 = G^2 N_a^2 N_p^{\text{tot}} = \sum_n \sigma^{-2} \quad (16)$$

per unit of $\langle J \rangle$. With the values used in the simulations ($G = 8.9 \times 10^{-7}$, $N_a = 2 \times 10^5$, $N_p^{\text{tot}} = 9.6 \times 10^8$) this gives $\text{SNR}_{\text{tot}}^2 \approx 3 \times 10^7$ and hence $P_{\text{sc}} \sim 10^2 / \text{OD}$, i.e. $\gamma_r T \sim 10^2 / \text{OD}$. For comparison the light-shift beam contributes $\gamma T = \beta T / \eta$, which for $T = 14 / \Omega$ and $\eta \sim 30$ is $\approx 0.5 \beta$. So unless $\text{OD} \gtrsim 10^3$ (a cavity, or a very dense cloud), the readout beam decoheres the atoms at least as strongly as the light-shift beam does for $\beta \lesssim 1$, and assumption A1 is not self-consistent with the assumed shot-noise level. Relaxing it costs one extra parameter, OD, and one change to the model: replace $\gamma \rightarrow \gamma + \gamma_r$ in (2) with $\gamma_r = P_{\text{sc}} / T$, independent of β . This also introduces the usual continuous-measurement trade-off between N_p and γ_r that the present model does not see. (In the single-beam scheme of Deutsch and Jessen the light-shift and readout beams are the same beam, so β , N_p and γ are all tied to one intensity.)

A2: no loss from the F manifold

The Raman process that gives (2) also transfers atoms to the other ground hyperfine manifold ($F = 4$ for Cs $F = 3$) with a branching ratio of order one, so loss occurs at a rate $\gamma_L \sim \gamma$. Within the F -manifold Hilbert space this is a non-trace-preserving term

$$\frac{d\rho}{dt} \supset -\frac{1}{2}\{K_L, \rho\}, \quad K_L = \sum_q L_q^{L\dagger} L_q^L = a + c J_z^2, \quad (17)$$

where $L_q^L \propto P_{F'} s_q P_F$ are the jump operators into F' and the form of K_L follows from z -symmetry and the $m \rightarrow -m$ reflection symmetry of π -polarized light. If the lost atoms are dark to the readout, the unnormalized $\text{Tr}[O\rho]$ is automatically the measured signal, and in the simplest uniform approximation $K_L = \gamma_L \mathbb{K}$ this is just a known factor $e^{-\gamma_L t}$ multiplying the signal and its derivatives. In the code this is one additional w -independent factor per step, so the fit and Fisher recursion are again unchanged. The complication that this approximation hides is that atoms in F' are not dark: they Faraday-rotate the readout with a different (roughly opposite) sign and evolve under a different light shift. Treating that properly requires enlarging the state space to $F \oplus F'$ (dimension 16 for Cs), which the superoperator scheme of Section 6 supports at the cost of a 256×256 dissipative map.

8 One beam: light shift, decoherence and readout from the same probe

In the actual scheme there is a single off-resonant, linearly polarized probe. It produces the tensor shift, it scatters photons, and its Faraday rotation is the readout. With photon flux Φ and detuning Δ , in the far-detuned limit,

$$\beta \propto \frac{\Phi \Delta'_{\text{HF}}}{\Delta^2}, \quad \gamma \propto \frac{\Phi \Gamma}{\Delta^2}, \quad G \propto \frac{1}{\Delta}, \quad \sigma^2 = \frac{1}{N_p G^2 N_a^2} \propto \frac{\Delta^2}{\Phi}, \quad (18)$$

where G is the rotation per atom per unit spin and $N_p \propto \Phi$ the photons per measurement interval δt . All four depend on Φ and Δ only through Φ/Δ^2 , so the probe strength is a single knob and two ratios are fixed by the physics:

$$\frac{\beta}{\gamma} = \eta, \quad \gamma \sigma^2 \delta t = \frac{1}{\text{OD}_{\text{eff}} N_a}, \quad \text{OD}_{\text{eff}} = c \frac{N_a \sigma_0}{A}, \quad (19)$$

with c an order-one factor from the vector polarizability and the spin-flip branching ratio. The second relation is the readout back-action of Section 7 made exact: the photons that carry the Faraday signal are the photons that scatter. Taking β as the sweep variable, the script `v6` derives per β

$$\gamma = \frac{\beta}{\eta}, \quad \sigma(\beta) = \sqrt{\frac{\eta}{\text{OD}_{\text{eff}} N_a \delta t \beta}}. \quad (20)$$

A stronger probe brings more nonlinearity, more decoherence and *less* shot noise; $\beta \rightarrow 0$ is no light at all, with no decoherence but no measurement either, so $\beta = 0$ is excluded from the sweep and there is no decoherence-free reference. The former constants G and N_p are absorbed into OD_{eff} ; with the values used in `v4/v5` and $\eta = 30$, $\text{OD}_{\text{eff}} = 10$, the old noise level $\sigma = 0.021$ corresponds to $\beta \approx 33$, a far stronger probe than any β in the `v5` sweep. The two physics inputs are now η (atom) and OD_{eff} (sample); both are placeholders in the launcher until derived and measured.

Expected shape of the optimum. The Fisher information per step is $(\partial \langle O \rangle / \partial w)^2 / \sigma^2 \propto \beta (\partial \langle O \rangle / \partial w)^2$, and the derivative decays on the scale η/β . For $T \ll \eta/\beta$ the information grows like βT^3 (more probe is better), for $T \gg \eta/\beta$ it saturates like η^3/β^2 (less probe is better), so the optimum sits at a finite $\beta_{\text{opt}} \sim \eta/T$ for every T . Unlike `v5`, where $\beta = 0$ eventually won because it was decoherence-free, the one-beam model always has a nonzero optimal probe strength.

Control experiment. Because the nonlinearity cannot be switched off physically without switching off the probe, `v6` has a flag `--tensor-scale` that multiplies the J_z^2 term in the Hamiltonian only, keeping the beam-derived γ and σ . Running the same sweep with `--tensor-scale 0` isolates what the nonlinear dynamics contributes to the estimation at fixed decoherence and fixed shot noise.